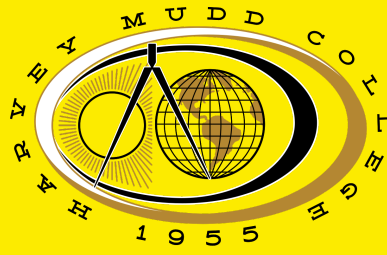


Cover

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Project Robocop Documentation



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Overview

Project Robocop is an ambitious keyboard restoration and modification program aimed at transforming a heavily worn yet structurally sound keyboard into an enthusiast grade input device. Inspired by the movie *Robocop*, this project seeks to replace and enhance key components to achieve an optimized and improved typing experience.

Upgrades

This project involved upgrading pre-existing or newly purchased components in order to augment the overall typing experience

- **Switches** – Each switch was disassembled, lubricated on 14 surfaces for smoothness, fitted with upgraded and lubricated springs, and filmed to improve stability
- **PCB** – The PCB had all 84 of its switches desoldered, was cleaned with isopropyl alcohol, and had specialized poron switch pads placed at each switch location to enhance sound and feel
- **Case** – Precisely cut thick foam inserts were placed throughout the case to minimize vibrations, creating a deeper and more satisfying sound profile

Total Cost

Below is an itemized table of all of the items purchased

Item	Quantity	Price	Vendor	Tax and Shipping
Panda MX Unlubed Switches	3	19.99	Glorious	6.15 + 10.99
GPL 105 Lubricant Oil	1	9.95	Thock King	1.95 + 3.95
120 pcs 0.5mm Poron Switch Pads	1	5.65	Thock King	
120 pcs 0.15 Polycarbonate Clear Switch Film	1	5.65	Thock King	
Lube Palette	1	3.99	Kinetic Labs	0.00 + 6.30
Tribosys 3203 Switch Lube	1	8.99	Kinetic Labs	
Magnifying Glasses	1	11.99	Kinetic Labs	0.00 + 10.84
Super Long Double Stage 63.5g Springs	1	13.89	Cannon Keys	1.29 + 4.60

4mm EVA Foam Sheet	1	11.49	Amazon	1.09 + 0.00
Switch Lubing Station and Kit	1	16.99	Amazon	1.61 + 0.00
Total: 189.50				

Starting Conditions

When I first bought the keyboard, it was in bad shape. It was in a heavily worn state with switches that felt inconsistent, scratchy, and sluggish while producing an unpleasant, hollow sound with every keystroke. The case amplified these issues, rattling with each press and lacking any kind of sound dampening, resulting in a loud, echoing, and unsatisfying typing experience. While the PCB and overall structure were intact and functional, the keyboard's feel and acoustics were far from acceptable, making it clear that a complete overhaul was necessary to transform it into a comfortable and satisfying daily-drivable device.

In Pursuit of Greatness

The main two problems were acoustics and feel. In order to eliminate or mitigate these issues, many aspects of the keyboard's construction had to change:

Feel – The feel of a keyboard refers to several qualities: the give of the PCB, the padding of its support, the quality of the switches in terms of smoothness and stability, the consistency of the stabilizers, and the stiffness of the overall case. As is my preference, the end goal revolves around the tactility of the keystroke, stiffness and lack of give in the PCB, and weight of the case.

Acoustics – The acoustics of a keyboard are the sum of many parts: the case material and design, the PCB and its thickness, any foam in the keyboard and its density, the switches and the lubrication, and any

modifications performed to the switches or PCB. In the end, the goal is a *full* sound profile that fills a complete spectrum of sounds with, as is my preference, a particular bias towards a lower and “thocky” sound signature.

Switch Modification

For each of the 108 switches, I performed no less than 6 major operations each with many sub-operations: opening, lubing, spring replacement, spring lubing, filming, and reassembly. By chunking the switches into groups of 36 and performing one operation on them, I could break the task into multiple sessions

PCB Modification

For each of the 84 slots on the PCB, I performed 4 major operations: desoldering, cleaning, padding, and re-soldering. These operations took less time than switch modification so I did them in individual sessions but completed them for all 84 slots

Case Modification

The interior of the case has an irregular shape, two switches and many screw standoffs that need to remain unobstructed, as well as the battery. In order to make the custom foam cutouts fit, I designed them to be like a jigsaw puzzle that could come together from many pieces

Milestones



Phase	Milestone	Operations
Phase 1	Keycap Removal	All 84 of the original deformed keycaps were removed using a specialized tool

Phase	Milestone	Operations
Phase 1	Primary Disassembly	The external and internal case screws were undone, the PCB and plate removed, and the battery disconnected
Phase 1	Desoldering	All 168 of the pins connected to the 84 switches were desoldered and then the switches removed using a specialized tool
Phase 2	Switch Opening	All 108 of the new switches were opened using a specialized tool
Phase 2	Switch Lubrication	A thin layer of Trybosys 3204 was brushed along the stem and bottom case of the switches
Phase 2	Spring Lubrication	The updated springs were bag lubed using GPL 105
Phase 2	Spring Replacement	The original springs were removed and then replaced with the new lubed ones using tweezers
Phase 2	Filming	Using tweezers, a switch film was placed on every switch

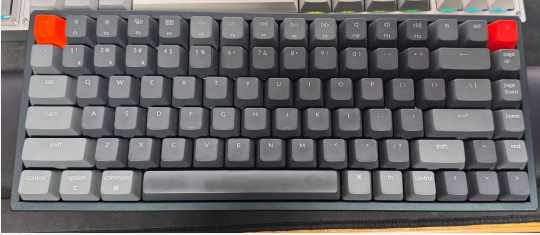
Phase	Milestone	Operations
Phase 2	Switch Reassembly	The switch top and bottom were put back together making sure the stem was in the correct orientation and the film was in the proper location
Phase 2	PCB Padding	Using tweezers, a poron switch pad was placed on every slot on the PCB. Some required modification to fit
Phase 2	Case Foam Measuring and Cutting	Segments of foam were cut out using scissors and razor blades and then placed in the case
Phase 3	PCB Reassembly	The switches were aligned with the PCB and fitted to the plate while making sure that all pins were undamaged
Phase 3	Soldering	The switches were soldered onto the PCB and then wiped down with isopropyl alcohol
Phase 3	Test Fitting	A series of test fits and modifications to the foam were done to ensure the side sliders, battery, and surface mounted PCB components fit

Phase	Milestone	Operations
Phase 3	Final Assembly	The battery was reconnected, sliders positioned, and screws tightened as it all came together
Phase 3	Testing	I then tested all of the switches, sliders, and diodes to ensure nothing was damaged
Phase 3	Keycap Installation	A new set of keycaps were installed and the keyboard was complete

Log

With reference to the Fabrication Milestones, this log reflects and documents the processes performed during the real project.

Phase 01



This is the starting condition of the keyboard. You can't see the majority of the issues but the keycaps are quite clearly warped and worn.

Starting Condition

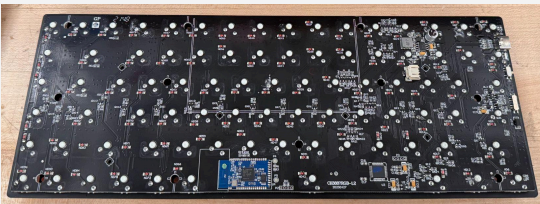
Phase 01



Showing the PCB and plate after cleaning but before removal of the original switches

Keycap Removal

Phase 01



PCB and Plate Removal

Showing the original empty space found in the case as well as the removed PCB and plate (top and bottom)

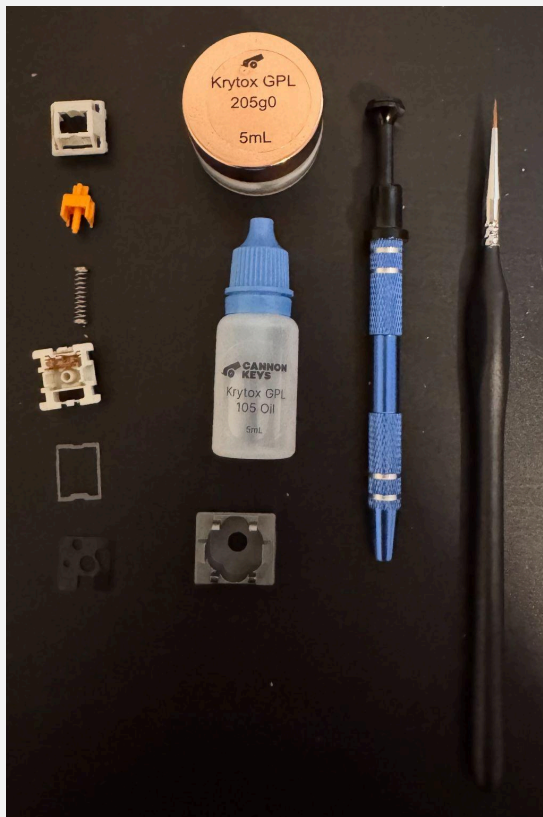
Phase 01



Showing the switches,
plate, and PCB after
desoldering and switch
removal

Desoldering and Switch Removal

Phase 02



This picture shows all the modifications done to the switches. The top four items in the leftmost column are the switch itself disassembled. Below it is the switch film and pad. To the right of that, from top to bottom, is the switch lube, spring lube, and switch opener. The blue item is a stem (the orange bit) holder and the last item is a lube brush

Switch Modifications

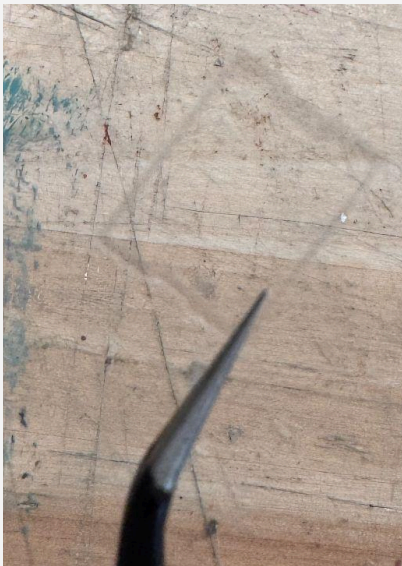
Phase 02



Showing actual process by which switches are lubricated. I disassemble a set of switches and placed their lowers, springs, and stems in racks while the tops were in a pile. I then lube the stem, lower, and springs and then reassemble the switches

Switch Lubrication

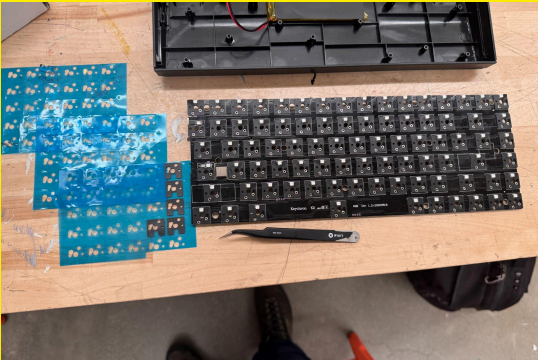
Phase 02



This shows the switch film held in curved needle nose tweezers. This is the best image I could get; the camera did not like focusing on a small clear object.

Switch Filming

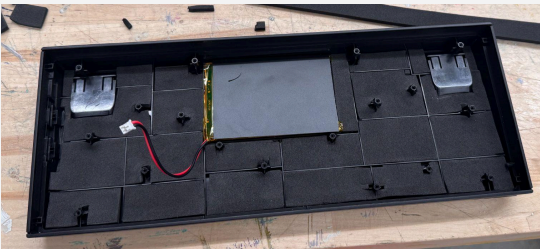
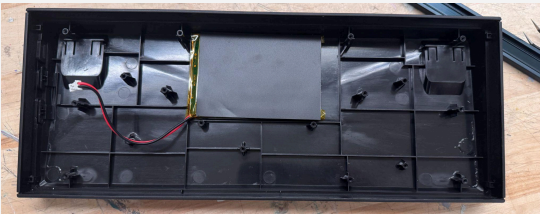
Phase 02



This shows the PCB with
the switch pads installed.

PCB Switch Pad Installation

Phase 02



These series of photos show the base state and then two layers of foam. The section on the left looks odd because there are a pair of sliders as well as some surface mount components that need space to function.



Case Foam Modification

Phase 03



This shows the switches soldered to the PCB with the plate as well.

PCB Assembly

Phase 03



This shows the keyboard fully assembled with the screws tightened and the battery connected.

Final Assembly

Phase 03



Project **Robocop**